

THE COURSE

Data Analysis & Machine Learning for Business

A practical course about working with data. Understanding matters more than remembering syntax.

Data Analysis & Machine Learning for Business

Andrey Shumilin

Six rules, and where everything lives.

01 Interrupt me

If something is unclear, stop me right there. No raised hands needed.

02 Just call me Andrei

No formalities, no patronymic.

03 Bring a laptop

We code live every session. Follow along, run it yourself, ask when it breaks.

04 Sign if I go too fast

Give me a sign and I will slow down or find different words.

05 Ask me anything about data

Any question about data science is welcome, even if it is not on today's plan.

06 I am not here to scare you

I am here to help you pick up practical skills you can use at work or in your daily routine.

ALL MATERIALS



github.com/seemsGoodNow/hse-data-science-course

BETWEEN SESSIONS

Ask in the **group chat**, not my DMs — someone has usually hit it already.

[@seems_good_now](#)

Ten minutes: join the chat, then tell me who you are.

1 Join the Telegram chat



t.me/+EJvuoZHhu7M2ZmZi

2 Fill in the survey



forms.gle/JchJimKWVagzocye8

I work with this kind of data every day.

NOW

DS at Sber (texts & agents)

I work on the Digital Assistant at Sber. My daily work is text analysis, building AI agents, AI system design

BEFORE

DS at Ozon (maths & operations)

E-commerce and warehouse data. These are the same processes this course is built on.

A TYPICAL DAY AT OZON

1. Translate a business question into a technical task
2. Collect the data from 10+ databases and 20+ tables
3. Find the insight, localize the problem, propose the fix

Data science just **changed**.

2022

Memorize hundreds of function signatures

Write almost everything by hand

Months of study before you could **build even a simple prototype**

Now

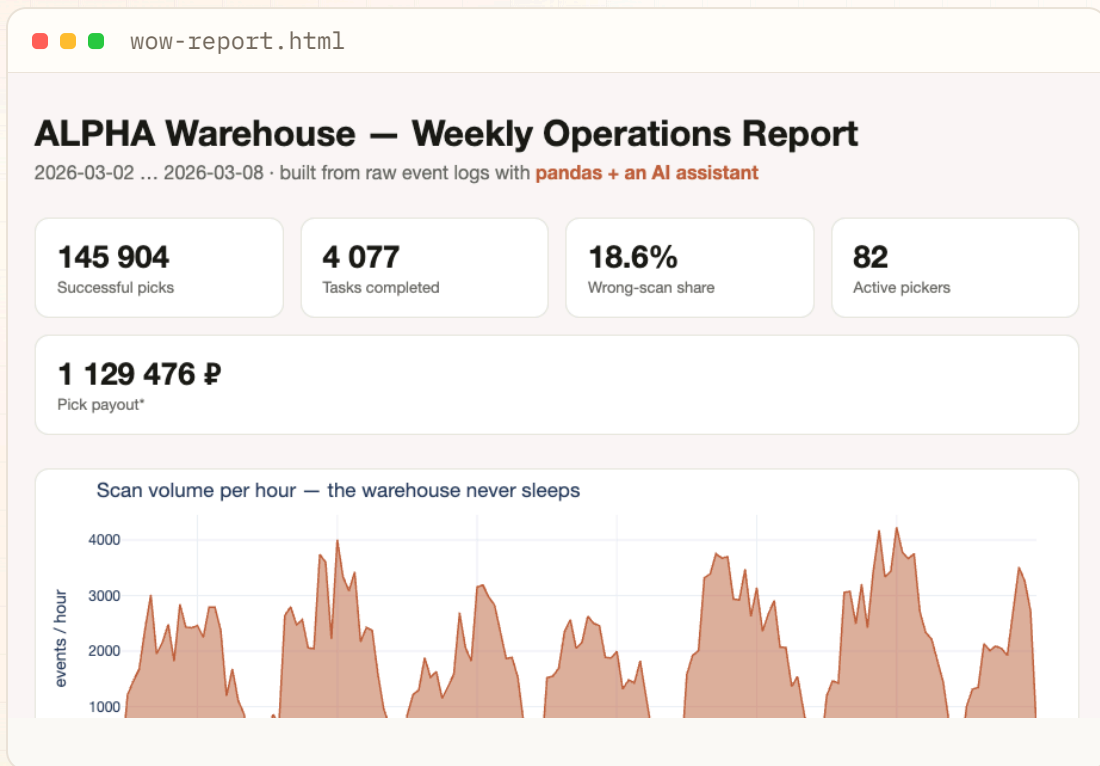
Think about what you actually need

Ask the AI clearly and precisely

Verify the result on the data

You learn fast, because AI explains as you go

The ideas and the checking stay with you. **The routine code can come from AI.**



From raw tables to this report.

An interactive report built with pandas and an AI assistant. You build it in the final session, and everything before that prepares you for it.

KPI cards hourly rhythm warehouse map

Something people actually open and read.

Seven sessions, two homeworks, one quiz.

01 Intro & where data comes from

today

02 Tables: filter, group, join

03 Charts & statistics

HW1

04 Hypothesis testing

05 Machine learning on tables

HW2

06 Model evaluation

Quiz

07 Clustering & the finale report

GRADING

40% Homework 1

40% Homework 2

20% Quiz

No exam and no project. Classes are in person.

Each homework is a small investigation on **your own slice** of data. Copying will not help you. AI is allowed, and you should be able to explain blocks of code.

The plan can shift: your survey answers and suggestions change what we spend time on.

GitHub for materials, Telegram for chat

Every business action leaves a **data trace**.

Nobody creates data for analysts. It appears as a by-product of normal operations.

One action leaves one or several rows.

THE ACTION	THE ROW IT LEAVES BEHIND
Pay by card	a transaction in the payments table
Order a taxi	a trip with pickup, route, price, driver
Like a song	an event in the recommender log
Scan a warehouse item	a row: time, worker, cell, item

To understand a table, understand the **process** that produced it.

A warehouse, in full detail.

Operations logs modelled closely on a real large e-commerce site. The processes, the noise and the errors behave like the real thing.

- 01
Truck arrives
- 02
Scanned into the system
- 03
Placed in a cell
- 04
Order comes in
- 05
Picking task created
- 06
Worker picks: container
→ cell → item
- 07
Container onto conveyor
- 08
Moves to consolidation
- 09
Item packed
- 10
Shipped → logistics

Thousands of rows a day, and all of it is **human movement**. Every scan along the way is one row with a timestamp, a worker, a cell and an item.

[Inside a real Ozon warehouse, read on Habr →](#)



Who touches data on the way to you.

SYSTEMS

Emit events

Apps & devices produce data as they run.

- Users make transactions
- Workers scan items

DATA ENGINEERS

Make it available

Collect it, store it, keep it clean.

...but sometimes there are no data engineers.

ANALYSTS / DS

Turn it into answers

Extract insight and build models.

- Propose the solution

← you are here

MANAGERS

Decide

Act on what the data says.

...or act without the data.

In finance you will often be the fourth box as well, so you need to **check what the third one hands you.**

Every business question is one of three types.

1 DESCRIPTIVE

What happened?

"Which zone is the slowest?"

Count, group and compare what is already in the table.

sessions 2–3

2 DIAGNOSTIC

Why did it happen?

"Why do pickers make more errors at night?"

Split the data, compare groups, check that the cause is real.

sessions 3–4

3 PREDICTIVE, ML

What will happen?

"How long will this task take?"

Use past data to estimate something that has not happened yet.

sessions 5–6

Most everyday value lives in the first two, in **counting and comparing things correctly**.

Excel and code each have their place.

Excel wins when...

- a quick look
- small data
- everyone already has it

Code wins when...

- 2 million rows are no problem
- repeatable: same report, new month, one command
- checkable line by line
- **AI speaks code natively**

One prompt can give you a whole script. Nothing gives you 40 correct right-clicks. **Code has become the easier path.**

Five tools, one sentence each.

**Py****Python** the language everything is written in**Jn****Jupyter notebook** a document where code, text and charts live together**VS****VS Code** the editor I use, and the standard professional setup**Cb****Google Colab** Jupyter in the browser, nothing to install**>_****Terminal** a text way to tell the computer what to do, and how you install things

Rule of the course: work where you feel comfortable. **Colab is always fine.**

Demo 01

Tool tour, from an empty terminal to your first notebook running in the browser.

- 1 Terminal: version check & `pip install pandas`
- 2 VS Code: a notebook with `2 + 2`, then an f-string
- 3 Open **the primer** in Colab, your homework notebook
- 4 The repo README: materials, primer, data dictionary

```
$ python --version
Python 3.13.5
$ pip install pandas
>>> 2 + 2
4
>>> f"margin = {price - cost}"
```

| PAUSE

Break 10 min

After the break: the Python speed-run. By 21:00 we open 312 thousand rows of a real-scale warehouse.

Colab did not open for you? Come up front now and we will fix it.

Six things, and you can read code.

01 Variables

```
revenue = price * qty
```

02 Lists & dicts

```
a list of dicts ≈ a table
```

03 if

```
make a decision
```

04 for

```
do it for every row
```

05 Functions

```
wrap it once, reuse it
```

06 Reading files

```
files in, structure out
```

Everything else is a library call or an **AI question** away.

Demo 02

Primer speed-run in the same notebook you opened in Demo 1. Show, predict, run.

- | | |
|---|---|
| 1 | §1 Cells, variables, types, f-strings (recap) |
| 2 | §2 Strings and the dot: <code>.strip().upper()</code> , dot+Tab |
| 3 | §3 Lists: <code>[0]</code> , <code>[-1]</code> , sum, max, len |
| 4 | §4 Dicts + list of dicts ≈ a table |
| 5 | §5–6 if and for: filter and count over orders |
| 6 | §7 Functions: wrap the margin calculation |
| 7 | §8 Read CSV + JSON + a pandas teaser |
| 8 | §9–10 Errors and asking AI. Notice: nothing memorised |

```
# what will this cell print?  
orders = [...]  
for o in orders:  
    if o["paid"]:  
        n += 1
```

Why the dot?

Every value in Python knows how to do a few things with itself. The dot asks it for one of them.

```
"hello".upper()
```

→ HELLO

```
orders.append(...)
```

→ the list grows by one

```
df.head()
```

→ the table shows its first rows

Type a dot and press **Tab** to see the menu of abilities.

Demo 03

First look at the warehouse, and the routine for meeting any unknown table.

- 1 Load and look: head, shape
- 2 dtypes, describe, value_counts
- 3 Events per day: find the two honest edges
- 4 **The AI moment: ask well, then verify**

```
# bad prompt
"my chart looks bad, fix it"

# good prompt
role, context, what I want, constraint
→ per-hour volume, day ticks "Mon 02"
```

Verify on the data: the rhythm repeats, the nights are not empty, and there is a dip at 9–11. **The chart cannot tell you why. The process owner can.**

The loop, every time.

STEP 1

Think

Try it yourself first, so you know what a good answer looks like.

STEP 2

Ask well

Role, context, what you have, what you want.

STEP 3

Verify

Always check the answer against the data.

WHICH TOOLS

DeepSeek, Qwen and GLM are free and good enough. I will demo **ChatGPT and Claude** in class.

THE RULE

Good at routine code, and risky where you cannot check it. **Explain every line you submit.**

Four things before next week.

1

Finish the primer

1-2 hours, solutions inside.

2

A working environment

Colab counts, that is enough.

3

Join the Telegram chat

All course communication lives there.

4

The survey, if you haven't

Two minutes; it shapes the pace.

Stuck for more than 10 minutes? The last section of the primer shows how to ask AI well. **If your setup is broken, message me.**

Five ideas to take home.

1 Data is a **by-product of processes**. Understand the process and you understand the table.

2 The **ideas and the checking are yours**. The routine code can come from AI.

3 **Six Python constructs** are enough to start.

4 The **dot** asks an object to use one of its abilities.

5 First-contact routine: `head` → `shape` → `dtypes` → `describe` → `value_counts` → one chart

NEXT WEEK deeper into tables: filter, group and join. This is where data starts answering questions, like which zone is slowest and whether the cause is the workers or the layout.